Combined Signal Representations for Modulation Classification Using Deep Learning: Ambiguity Function, Constellation Diagram, and Eye Diagram

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2023 32nd Wireless and Optical Communications Conference (WOCC) | 979-8-3503-3715-0/23/$31.00 ©2023 IEEE | DOI: 10.1109/WOCC58016.2023.10139474

***Abstract*—We exploit deep learning convolutional neural networks (CNN) based on joint image representation and propose an automatic modulation classification algorithm to classify the communication signals. The combined representations include a constellation diagram, an ambiguity function (AF), and an eye diagram. Experimentation results show that combining constellation and eye diagrams achieves superior classification performance compared to having these representations separately. Combining AF and an eye diagram results in improvement at low SNR.**

***Index Terms*—Spectrum awareness, deep learning, modulation classification, constellation diagram, eye diagram, ambiguity function, quadrature amplitude modulation (QAM).**

I. Introduction

Automatic modulation classification (AMC) has various applications for civilian, military, and electronic warfare. Spectrum awareness is becoming more important today due to the evolution and rapid expansion of radio communications. Because of that, automatic signal recognition and modulation classification applications are becoming more important [1], [2].

In this paper, we consider the deep learning-based modulation classification that utilizes a joint image combined representation as the input for the convolution neural network (CNN), as shown in Fig 1. Deep learning has shown excellent performance in computer vision [3]–[7]. To exploit that, we convert the received signal into an image and use CNN to classify it [3].

There are some unique and important image-based signal representations used in wireless communications, such as constellation diagrams, eye diagrams, and ambiguity functions. A constellation diagram is a widely used 2D signal representation that maps the modulated signal into scattering points on a plane. The scattering points are created by digital modulation schemes like

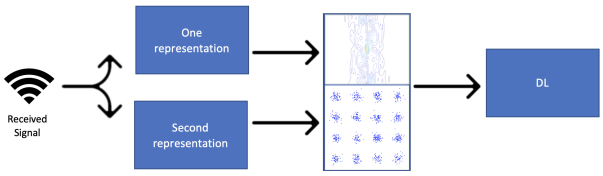


Fig. 1. Creating a combined signal representation.

quadrature amplitude modulation (QAM) or phase shift keying (PSK). QAM uses amplitude and phase variations to achieve high levels of spectrum usage efficiency [8]. Clustering and regression on constellation signatures are used for modulation classification of blind signals [9]. While in [10], Machine Learning (ML) is used to classify constellation diagrams with different fades. In [11], a color image is generated and classified using constellation density. And in [2], QAM classification is used to estimate the symbol rate and phase offset of the modulation.

One important signal representation, in addition to the constellation diagram, is the eye diagram. In practical applications, the eye diagram is used to gauge device performance by testing and measuring the digital signal in the time domain. Engineers also use it to obtain a quick evaluation of system performance and gain information about the channel [12]. Some recent work use eye diagrams as stand-alone input with deep learning for automatic modulation classification [13], [14].

The ambiguity function (AF) is a major tool for analyzing radar signals. It's the third signal representation considered in this paper. The ambiguity function describes two dimensions of the signal, delay and Doppler. An ideal ambiguity function that shows a high magnitude exports at the middle of the figure. This is the reason why it is used for time-frequency analysis. Like in

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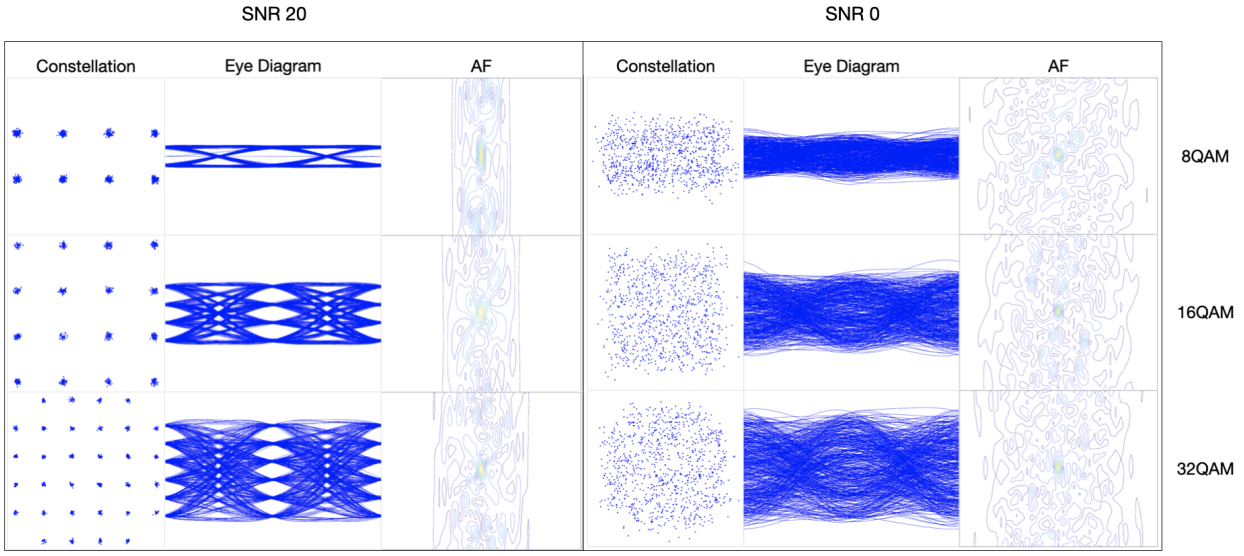


Fig. 2. Signal representation in AF, constellation, and eye diagrams at SNR 20 and 0 dB.

[15], AF is used to classify QAM PSK, amplitude-shift keying (ASK), frequency-shift-keying (FSK), minimum-shift keying(MSK), linear frequency modulation (LFM), and orthogonal frequency division multiplexing (OFDM) modulations.

Combined signal representations were utilized in the literature. As in [16], the received signal is converted into a time–frequency image and an instantaneous auto-correlation map. The two images are converted to binary and combined in vertical order. The upper half part is an instantaneous auto-correlation image and the lower half is time–frequency image. The result is used as an input of CNN to classify the modulation of radar emitter signals. While in [17], Cyclic spectra image and constellation diagrams were used to leverage the strong noise resistibility of the first image and the high-order modulation capability of the second one. The images were fed into a two-branched CNN.

This paper is organized as follows. Section II describes the methods used in data generation and deep learning. The results and discussions are presented in Section III. Finally, the paper is concluded in Section IV.

II. Methods and Data Generation

In this section, we present our design and experiments using deep learning for the combined signal representations. Our research is built based on two parts. First, signals are generated for three signal representations separately with different noise levels. Then we pair every two representations together. Second, we use our model to classify the generated signals.

A. Generating Signals

For each signal representation (constellation diagram, eye diagram, and AF), we created data sets with different noise levels ranging from SNR -20 dB to 10 dB: (-20, -15, -10, -5, 0, 5, 10).

In all the data sets, we generated the signal with 8QAM, 16QAM, and 32QAM modulations. We picked these three modulations as they are commonly used in the field of signal modulation. These signals are considered under Gaussian noise, as in Fig. 2.

We use 350 samples for training and 150 for testing in each signal representation at each different SNR. For the eye diagram, we used this configuration for setting up a raised cosine transmit filter to obtain a realistic eye diagram:

- Filter shape = Normal
- Roll-off factor = 0.85
- Filter span in symbols = 8
- Output samples per symbol = 24
- Linear filter gain = 1

For our ambiguity function, we use `rcosdesign` function (Raised cosine FIR pulse-shaping filter design), which takes three parameters:

- Rolloff factor = 1
- Number of symbols = 20
- Samples per symbol = 1
- Shape of the raised cosine window = `sqrt`

The rolloff factor determines the bandwidth of the filter. From zero to one, zero corresponds to a brick-wall filter, while one means pure raised cosine.

When combining signal representation, we add one representation on top of the other vertically, as shown in Fig 1.

B. CNN Model

Our CNN model is fast to compute, going seven layers deep, as shown in Fig. 3. With three epochs to train the

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data and Adam function `optim.Adam` as optimizer. The algorithm uses two convolutional layers and two pooling layers, then goes through three fully connected hidden layers. The model overall structure is CONV/RELU/POOL/CONV/RELU/POOL/FC/RELU/FC.

The first convolutional layer has three input channels, six output channels, a kernel size of 3 to form a 3x3 filter, and a step size of 1. The second layer takes the six features that got out from the first layer and then outputs 16 channels for feature extraction. The fully connected layers have an input size that deepens on the size of the image multiplied by 16, the output of the second convolution layer. The first layer outputs 120, the second 84, and the third outputs three since we classify three classes.

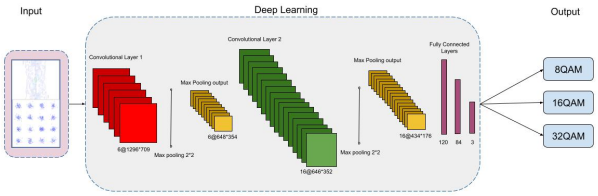


Fig. 3. Our CNN model.

III. Results and Discussions

In this section, we present the experiment results and discussions. First, classify the signal representations solely with the impact of noise. Then we combine the representations in the second part of the testing.

A. Impact of Signal-to-Noise Ratio

For each separate signal representation, we first classify the signal with low noise (10 SNR). Then we increment the noise by 5 dB SNR until -20 dB SNR and see if we can distinguish the different QAM modulations along the way. As we can see from Fig. 4 (A) eye diagram and constellation diagram did much better than the ambiguity function. At 10 dB SNR, all three achieve near 100%. At 5 dB SNR, we start to see the gap among constellation, eye diagrams, and AF. The first two stay at 100% while AF drops to 56%. Going through 0 dB SNR, we see the same trend, constellation and eye diagrams achieve 100% while AF accuracy drops to 33%. AF stays at 33% till the end of the experiment (-20 dB SNR). At the same time, the other two (constellation and eye Diagram) achieve high accuracy at -5 dB SNR: 95% and 100%, respectively. At -10 dB SNR, Accuracy is still high, with 99% and 91% in the same order. While at -15 dB SNR, the accuracy takes a hit with 52% for constellation and 33% for Eye diagram. Lastly, for -20 SNR, we get 42% for the constellation and 33% for the eye diagram.

B. Impact of Combining Signal Representations

In the second part of the experiment, we combined the signal representations. Then feed it into our CNN model.

1) Combining of the Constellation Diagram and Eye Diagram: As we can see in Fig 4 (B), combining the constellation diagram and eye diagram leads to better classification results. We can see that clearly at -15 dB, where the combined accuracy is 62%, while the eye diagram was 33%, and the constellation diagram was at 52%.

2) Combining Ambiguity Function and Eye Diagram: We see a similar trend in Fig 4 (C). Adding AF and eye diagrams together improves the results. Especially at -15 dB, where it improves from 33% accuracy for both AF and eye diagram to 37% with them combined. The combined representation of the rest of the figure looks similar to the eye diagram results.

3) Combining Constellation Diagram and Ambiguity Function: While in Fig 4 (D), the combined representations did not improve the accuracy. Having both AF and constellation diagram representation together did not improve the results from having a constellation diagram solely. That is mainly because we added the best signal representation (constellation diagram) with the worst signal representation in our test (AF).

We ran the experiments using Mac-Book pro 2019, with 2.3 GHz 8-Core Intel Core i9 and 16 GB 2400 MHz DDR4 memory.

IV. Conclusions

Our CNN model showed which signal representation is better to classify out of those being tested. The constellation diagram got better accuracy compared to the eye diagram and AF. The paper also showed that when combining signal representations, specifically constellation and eye diagrams, we got better results than having them separately. Combining AF and an eye diagram improved the signal slightly at -15 dB. While having a constellation diagram with AF did not improve the results.

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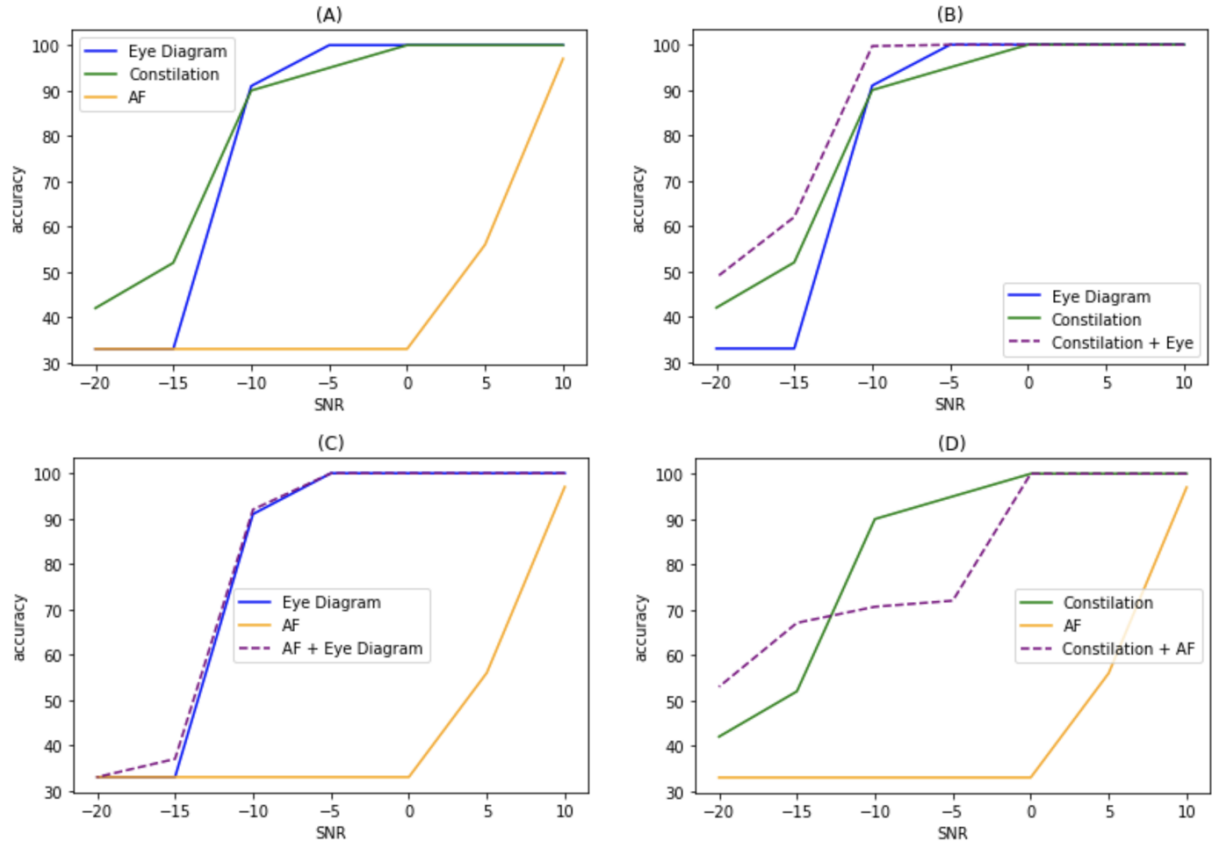


Fig. 4. The experiment results (A) The three representations separately. (B) Combining the constellation diagram and eye diagram. (C) Combining ambiguity function and eye diagram. (D) Combining constellation diagram and ambiguity function.

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